Problem 1:

$12_{10} \& 7_{10}$

$= 1100_2 \& 0111_2$

$= 0100_2$

$= 4_{10}$

Problem 2:

$15_{10} | 10_{10}$ $\quad$ $15_{10} | 10_{10}$

$= 1111_2 | 1010_2$

$= 1111_2$

$= 15_2$ $\quad$ 15_{10}

$$\begin{array}{r} 1111 \\ | \;\; 1010 \\ \hline 1111 \end{array}$$

Problem 3:

$20_{10} \wedge 13_{10}$

$= 10100_2 \wedge 01101_2$

$= 11001_2$

$= 25_{10}$

Problem 4:

$\sim 10_{10} = \sim(00001010_2) = 11110101_2$

$= -128+64+32+16+0+4+0+1$

$= -11$

(2's complement form for -ve numbers)

$= 2\text{'s complement}(10_{10})$

$= 2\text{'s complement}(00001010)_2$

$= 1\text{'s complement}(11110101)_2 + 1_2$

$= 11110110_2$

$= 1\text{'s complement}(00001010_2) + 1_2$

$= (11110101)_2 + 1_2$

$= 11110110_2$

$= -128+64+32+16+0+4+2+0$

$= -10$

HW

Problem 5:

$8_{10} << 2$

$= 1000_2 << 2$

$= 100000_2$

$= 32_{10}$

$x << k$

$= x * 2^k$

k	$x << k$
0	x
1	$x \times 2$
2	$x \times 4$
3	$x \times 8$
$\vdots$	
k	$x \times 2^k$

Problem 6:

$64_{10} >> 3$

$= 1000\,000_2 >> 3$

$= 1000_2$

$= 8_{10}$

$64_{10} >> 3$

$x >> k$

$= x // 2^k$

$\neq \frac{x}{2^k}$

k	$x >> k$
0	x
1	

$x = 13 \quad k = 2$

$x >> k = 13 >> 2$

$= 1101_2 >> 2$

$= 11_2 = 3$

$\frac{x}{2^k} = \frac{13}{4} = 3.25$

33

$>> 4$

16

100001

Problem 7:

$$(5_{10} \ \& \ 3_{10}) \mid 4_{10}$$

$$= (101_2 \ \& \ 011_2) \mid 100_2$$

$$= 001_2 \mid 100_2$$

$$= 101_2$$

$$= 5_{10}$$

✓

Problem 8:

$(10_{10} >> 1) \& 7_{10}$

$= (1010_2 >> 1) \& 111_2$

$= 101_2 \& 111_2$

$= 101_2$

$= 5_{10}$

Problem 9:

$1 \wedge 1 \wedge 1$

XOR $\oplus$ is associative

$= 1 \wedge (1 \wedge 1)$

$= 1 \wedge 0$

$= 1$ ✓

Problem 10:

Comp: complement

$\sim(-5_{10})$

$\text{not}(x) = -x-1$

$\sim(-5) = -(-5)-1 = 5-1 = 4$

$= \sim(2\text{'s comp}(5_{10}))$

$= \sim(2\text{'s comp}(00000101_2))$

$= \sim(1\text{'s comp}(00000101)+1)$

$= \sim(11111011_2) \quad = \sim(11111010_2 + 1_2)$

$= 00000100_2$

$= 4_{10}$

Problem 11: {T: True, F: False

T or F and F

= T or (F and F)

= T or F

= T

Problem 12:

not (T and F)

= not(F) not (F)

= T

Problem 13:

$(5>3)$ and $(2==2)$

$=$ T and $(2==2)$

$=$ T and T

$=$ T

Problem 14:

0 or 5

= F or T

= T (Because in python, 0 = F and numbers >= 1 = T)

<= -1 are also T

(Python is giving 5. why?)

If the first operand is false for OR, result = second operand

If the first operand is true for OR, result = first operand

Problem 15:

10 and 20

= T and T

= T   (Because in python, 0 = F and numbers >= 1 = T)
<= -1 are also T

(Python is giving 20. why?)

For 'and', if first operand is:
(1) True ($\neq 0$) then result = second operand
(2) False (= 0) then result = 0.

Problem 16:

not ("apple" == "orange")

= not (F)

= T

Problem 17:

$a = T, \; b = T, \; a \mathrel{!}= b$

$a \mathrel{!}= b$

$\Rightarrow T \mathrel{!}= T$

$= F$

Problem 18:

Left circular shift on 128_{10} by 1 bit

128_{10} circular << 1

$= 10000000_2$ circular<< shift

$= 00000001_2 = 1$

$= 01000000_2$

$= 64_{10}$

$n = 00abcdef_2$

$n \overset{circ}{<<} 6$

$A = (abcdef000000_2) \% (100000000_2)$

$= ef000000$

$B = 00abcdef_2 >> 2 = 00abcd_2$

$A|B = ef00abcd_2$

Eg: $101_2 \overset{circ}{<<} 6$ bits

$N = 8$ (total bits)

$(00000101_2) \overset{circ}{<<} 6$

$(n<<d) \% (1<<N)$

rotated left part $A = (00000101000000)_2$
$\% 100000000$

$= 01000000_2$ (last 8 bits)

rotated right part $B = (n >> (N-d))$

$= 101_2 >> (8-6)$

$= 101_2 >> 2 = 1_2$

ans: $A|B = 01000001$

Problem 19:

$\text{bool}(C10) \wedge \text{bool}(C0)$

$= T \wedge \text{bool}(C0)$

$= T \wedge F$

$= T$

✓

Problem 20:

$(7_{10}\ \&\ 1) == 1$ and $(8_{10} >> 1) == 4_{10}$

$\Rightarrow (111_2\ \&\ 1) == 1$ and $(1000_2 >> 1) == 100_2$

$111_2\ \&\ 001_2$

$\Rightarrow 1 == 1$ and $(1000_2 >> 1) == 100_2$

100_2

$=$ T and T

$=$ T

✓

Problem 21:

T or F and not T

= T or (F and not T)

= T or (F and F)

= T or F

= T

Problem 22:

$(23 > 42)$ and $(36 > 12)$ or $(39 <= 39)$

$=$ F and $(36 > 12)$ or $(39 <= 39)$

$=$ F and T or $(39 <= 39)$

$=$ F and T or T

$=$ F or T

$=$ T

Problem 23:

$10_{10} + 5_{10} << 1$

$= (10+5)_{10} << 1$

$= 15_{10} << 1$

$= 1111_2 << 1$

$= 11110_2$

$= 30_{10}$

Problem 24:

A or (not A and B)

= (A or not A) and (A or B)

= True and (A or B)

= A or B

Problem 25:

(A and B) or (A and not B)

= A and (B or not B)

= A and T

= A

Problem 26:

A or (F and B)

= A or F

= A

Problem 27:

not(not A or B)

= ~~(not(not A) and~~

= (not (not A)) and (not B)

= A and not B

Problem 28:

$$5_{10} \times 2^2_{10} \;\&\; 15_{10}$$

$$= (5_{10} \times (2^2_{10}))\;\&\; 15_{10}$$

$$= (5_{10} \times 4_{10})\;\&\; 15_{10}$$

$$= 20_{10} \;\&\; 15_{10}$$

$$= 10100_2 \;\&\; 01111_2$$

$$= 00100_2$$

$$= 4_{10}$$

Problem 29:

$$\text{not}(15//3==5 \text{ and } 7\%2==0)$$

$$= \text{not}(5==5 \text{ and } 1==0)$$

$$= \text{not}(T \text{ and } F)$$

$$= \text{not}(F)$$

$$= T$$

Problem 30:

(A or B) and (not A or B)

= (A and not A) or B

= F or B

= B

Problem 31:

$(A \text{ and } B \text{ and } C) \text{ or } (A \text{ and } B \text{ and not} C)$

$= A \text{ and } B \text{ and } (C \text{ or not} C)$

$= A \text{ and } B \text{ and } T$

$= A \text{ and} B$

✓

Problem 32:

$\sim 5_{10}$ & 7

$= \sim 101_2$ & 111_2

$= 010_2$ & 111_2

$= 10_2$

$= 2$

$= \sim(00000101_2)$ & (111_2)

$= (11111010)_2$ & (00000111_2)

$= (00000010)_2$

$= 2$

$\sim 5_{10} \mid 7_{10}$

111_2

$= 7$

with 2's complement

$(11111010)_2 \mid (00000111)_2$

$= 11111111_2$

$= -128+64+32+16+8+4+2+1$

$= -1$

Problem 33:

$1 << 3 \mid 1 << 1$

$1000_2 \mid 0010_2$

$= 1010_2$

$= 10_{10}$

Problem 34:

A and (a or b)

= a (absorption)

A

Problem 35:

(T or F) and (not T or T)

= T and (not T or T)

= T and (F or T)

= T and T

= T

Problem 36:

$$A \text{ or } (A \text{ and } B) \text{ or } (B \overset{\text{and}}{\cancel{\text{or}}} F)$$

$$= A \text{ or } (B \overset{\text{and}}{\cancel{\text{or}}} F) \qquad \text{(Absorption)}$$

$$= A \text{ or } \underset{F}{\cancel{B}}$$

$$= A \text{ or } F$$

$$= A$$

problem 37:

1_{10} 8-bit circular >>1

$= 00000001_2$ circular >>1

$= 10000000_2$

$= 256_{10}$ 128_{10}

X

Problem 38:

$10 > 5 > 2$

$= (10 > 5) \text{ and } (5 > 2)$

$= T \text{ and } T$

$= T$

Problem 39:

not(5<2) or 10//2==5

=not (F) or 10//2==5

= T or 10//2==5

= T or 5==5

= T or T

= T

Problem 40:

(A and B) or C = A or C & B or C

(A and B) or (not (A and B))

= (A and B) or (not A or not B)

= ((A and B) or not A) or not B

= ((A or not A) and (B or not A)) or not B

= ((T) and (B or not A)) or not B

= (B or not A) or not B

=

= (B or not A) or not B

= (B or not B) or (not A) [associative]

= T or not A

= T

Let A and B = X.

(A and B) or (not (A and B))

= X or not X

= T